

Invasive Plants in Pennsylvania

Starry stonewort

Nitellopsis obtusa



NYS Dept. of Environmental Conservation

Background/Range:

This species is native to Europe and parts of Asia. It was first introduced to North America in the 1970s along the St. Lawrence River and is now present throughout the Great Lakes states and parts of the Northeast and Canada.

Habitat:

In the Great Lakes region, this species can be found in relatively calm, deep waters (4-30 meters) often near marinas and boat launches.



Photo by Paul Skawinski

Description:

Starry stonewort is a freshwater microalga (micro-algae). Its bright-green branchlets are attached to the stem-like central axis in whorls of 4-8. The central axis (main stem) can be over 2 meters long, with branchlets reaching several inches. It is anchored to the substrate by root-like rhizoids. This species can be distinguished from the other carophytes by the presence of asexual reproductive structures, called bulbils, that are white, star-shaped structures about the size of a grain of rice.



Starry stonewort bulbil by Paul Skawinski

Biology and Spread:

Presently, there has been no documented cases of female plants in the United States. Therefore, there is no seed production taking place. The transport of stem fragments and bulbils are the only methods of reproduction. The male plants produce small, orange spheres called antheridia which appear at the branchlet nodes.

Ecological Threat:

Once established, this species can form dense mats at the water's surface which can outcompete native vegetation, reduce fish spawning habitat and impact water-related recreation.

How to Control this Species:

Due to the limited success of current control methods, prevention of new infestations and education of water recreationists should be considered the best control options. Mechanical control is extremely difficult and should only be attempted on small or recently introduced populations. It involves hand-pulling or another type of aquatic vegetation harvesting equipment to remove all plant material.

Chemical treatments must be undertaken with extreme care to avoid impacts to other aquatic resources in the waterbody. Limited control can be achieved using a copper-based algaecide, while the plant is still low growing. However, as plant height increases the lower portions are buffered from uptake and will survive.



Photo by Paul Skawinski